

# Passive Infrastructure Identification

[Netcraft](#) can offer us information about the servers without even interacting with them, and this is something valuable from a passive information gathering point of view. We can use the service by visiting <https://sitereport.netcraft.com> and entering the target domain.

The screenshot shows the Netcraft website interface. The browser address bar displays the URL `https://sitereport.netcraft.com/?url=https%3A%2F%2Ffacebook.com`. The page title is "Site report for https://facebook.com". Below the title, there is a search bar with the text "Look up another site?". The page is divided into two main sections: "Background" and "Network".

**Background**

|                  |                                                                                                   |                      |          |
|------------------|---------------------------------------------------------------------------------------------------|----------------------|----------|
| Site title       | Facebook - log in or sign up                                                                      | Date first seen      | May 1997 |
| Site rank        | 5641                                                                                              | Netcraft Risk Rating | 0/10     |
| Description      | Log in to Facebook to start sharing and connecting with your friends, family and people you know. |                      |          |
| Primary language | English                                                                                           |                      |          |

**Network**

|                         |                                                         |                         |                                                                  |
|-------------------------|---------------------------------------------------------|-------------------------|------------------------------------------------------------------|
| Site                    | <a href="https://facebook.com">https://facebook.com</a> | Domain                  | <a href="https://facebook.com">facebook.com</a>                  |
| Netblock Owner          | Facebook, Inc.                                          | Nameserver              | a.ns.facebook.com                                                |
| Hosting company         | Facebook                                                | Domain registrar        | registrarsafe.com                                                |
| Hosting country         | US                                                      | Nameserver organisation | whois.registrarsafe.com                                          |
| IPv4 address            | 157.240.221.35 (VirusTotal)                             | Organisation            | Facebook, Inc., 1601 Willow Rd, Menlo Park, 94025, United States |
| IPv4 autonomous systems | AS32934                                                 | DNS admin               | dns@facebook.com                                                 |
| IPv6 address            | 2a03:2880:f158:82:face:b00c:025de                       | Top Level Domain        | Commercial entities (.com)                                       |
| IPv6 autonomous systems | AS32934                                                 | DNS Security Extensions | unknown                                                          |
| Reverse DNS             | edge-star-mini-shv-01-lhr8.facebook.com                 |                         |                                                                  |

Some interesting details we can observe from the report are:

|                 |                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------------------|
| Background      | General information about the domain, including the date it was first seen by Netcraft crawlers. |
| Network         | Information about the netblock owner, hosting company, nameservers, etc.                         |
| Hosting history | Latest IPs used, webserver, and target OS.                                                       |

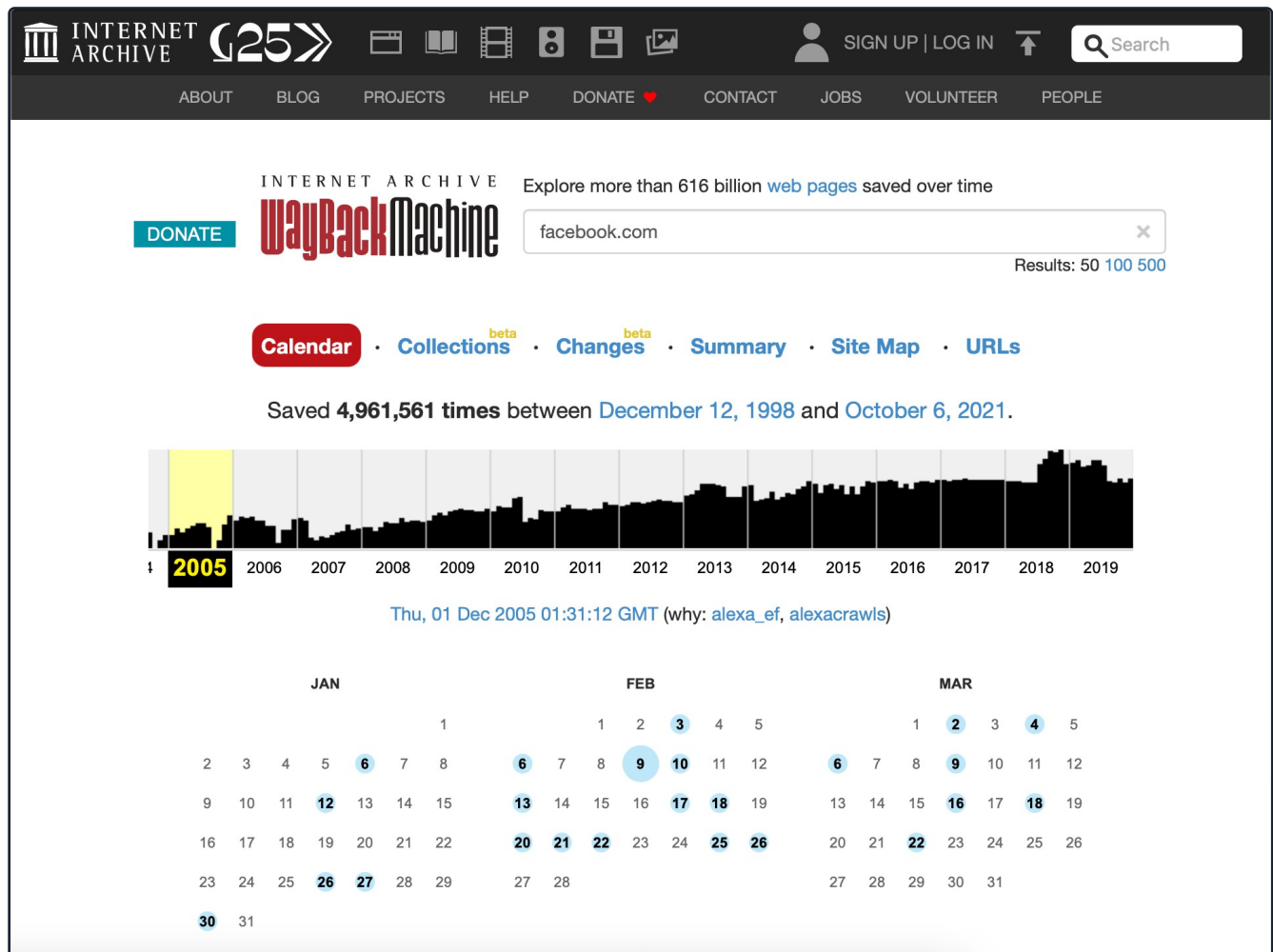
We need to pay special attention to the latest IPs used. Sometimes we can spot the actual IP address from the webserver before it was placed behind a load balancer, web application firewall, or IDS, allowing us to connect directly to it if the configuration allows it. This kind of technology could interfere with or alter our future testing activities.

## Wayback Machine

The [Internet Archive](#) is an American digital library that provides free public access to digitalized materials, including websites, collected automatically via its web crawlers.

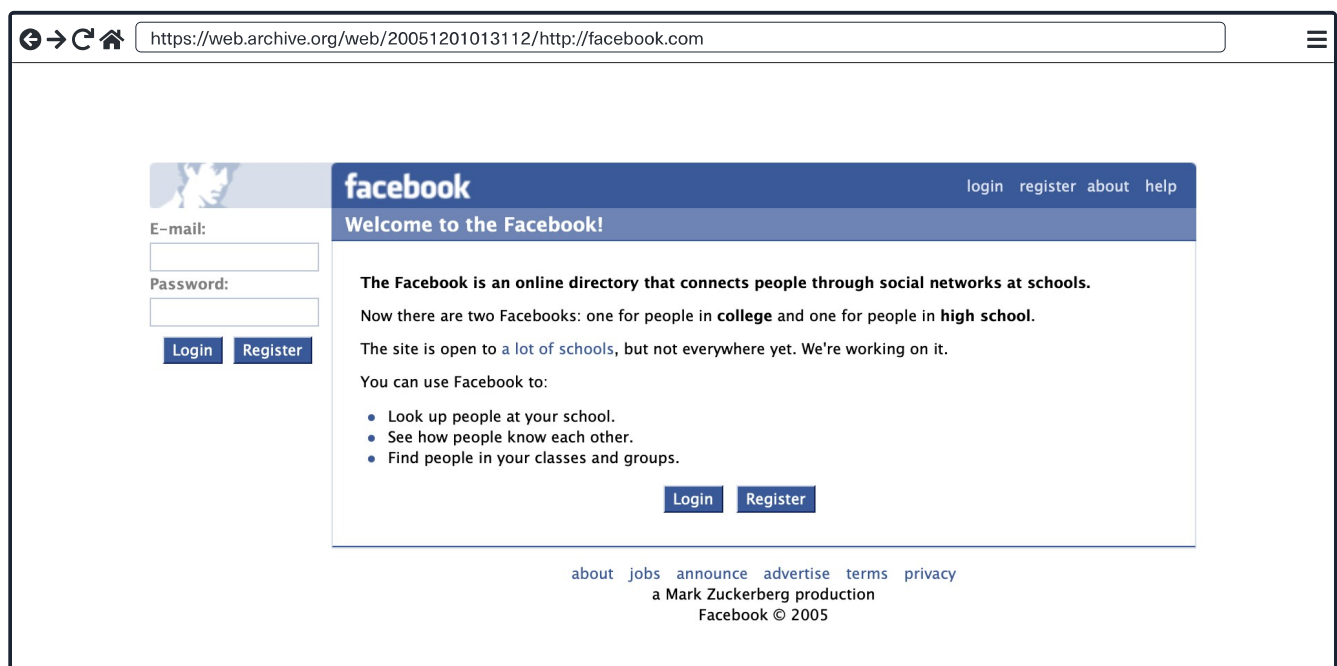
We can access several versions of these websites using the [Wayback Machine](#) to find old versions that may have interesting comments in the source code or files that should not be there. This tool can be used to find older versions of a website at a point in time. Let's take a website running WordPress, for example. We may not find anything interesting while assessing it using manual methods and automated tools, so we search for it using Wayback Machine and find a version that utilizes a specific (now vulnerable)

plugin. Heading back to the current version of the site, we find that the plugin was not removed properly and can still be accessed via the `wp-content` directory. We can then utilize it to gain remote code execution on the host and a nice bounty.



The image shows the Wayback Machine interface for the URL `facebook.com`. The top navigation bar includes links for ABOUT, BLOG, PROJECTS, HELP, DONATE, CONTACT, JOBS, VOLUNTEER, and PEOPLE. The main header features the Internet Archive logo and the text "Explore more than 616 billion web pages saved over time". Below this, a search bar contains the URL `facebook.com`, and the results show "Results: 50 100 500". The interface includes a "Calendar" button and links for "Collections", "Changes", "Summary", "Site Map", and "URLs". A message states "Saved 4,961,561 times between December 12, 1998 and October 6, 2021." A bar chart shows the frequency of captures over time, with a highlighted yellow bar for the year 2005. Below the chart, a calendar view for December 2005 is shown, with the date "Thu, 01 Dec 2005 01:31:12 GMT" selected. The calendar also shows the months of January and February.

We can check one of the first versions of `facebook.com` captured on December 1, 2005, which is interesting, perhaps gives us a sense of nostalgia but is also extremely useful for us as security researchers.



The image is a screenshot of the archived Facebook homepage from December 1, 2005. The browser address bar shows the URL `https://web.archive.org/web/20051201013112/http://facebook.com`. The page features the Facebook logo and a navigation bar with links for "login", "register", "about", and "help". The main content area includes a "Welcome to the Facebook!" message and a description of the site as an online directory connecting people through social networks at schools. It mentions that there are two Facebooks: one for people in college and one for people in high school. The site is open to a lot of schools, but not everywhere yet. Below this, a list of features is provided: "Look up people at your school", "See how people know each other", and "Find people in your classes and groups". At the bottom, there are "Login" and "Register" buttons. The footer contains links for "about", "jobs", "announce", "advertise", "terms", and "privacy", along with the text "a Mark Zuckerberg production" and "Facebook © 2005".

We can also use the tool `waybackurls` to inspect URLs saved by Wayback Machine and look for specific keywords. Provided we have `Go` set up correctly on our host, we can install the tool as follows:

```
dbcyp0n@htb[/htb]$ go install github.com/tomnomnom/waybackurls@latest
```

To get a list of crawled URLs from a domain with the date it was obtained, we can add the `-dates` switch to our command as follows:

```
dbcyp0n@htb[/htb]$ waybackurls -dates https://facebook.com > waybackurls.txt
dbcyp0n@htb[/htb]$ cat waybackurls.txt

2018-05-20T09:46:07Z http://www.facebook.com./
2018-05-20T10:07:12Z https://www.facebook.com/
2018-05-20T10:18:51Z http://www.facebook.com/#!/pages/Welcome-Baby/143392015698061?ref=tsrobots.txt
2018-05-20T10:19:19Z http://www.facebook.com/
2018-05-20T16:00:13Z http://facebook.com
2018-05-21T22:12:55Z https://www.facebook.com
2018-05-22T15:14:09Z http://www.facebook.com
2018-05-22T17:34:48Z http://www.facebook.com/#!/Syerah?v=info&ref=profile/robots.txt
2018-05-23T11:03:47Z http://www.facebook.com/#!/Bin595

<SNIP>
```

If we want to access a specific resource, we need to place the URL in the search menu and navigate to the date when the snapshot was created. As stated previously, Wayback Machine can be a handy tool and should not be overlooked. It can very likely lead to us discovering forgotten assets, pages, etc., which can lead to discovering a flaw.

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### My Workstation

O F F L I N E

▶ Start Instance

∞ / 1 spawns left